

Multiple Choice

1. **Answer:** D

Options: A) O₂; B) CH₄ (Methane); C) CO₂; D) H₂O; E) N₂ (Nitrogen)

Note: Correct option: D - H₂O

2. **Answer:** A

Options: A) The human genome has fully adapted to current environmental pressures; B) Modern medicine has eliminated the effects of genetic drift; C) The rate of genetic drift is directly proportional to technological advancements; D) Humans have reached a genetic equilibrium where drift is no longer possible; E) Humans have developed immunity to genetic drift

Note: Correct option: A - The human genome has fully adapted to current environmental pressures

3. **Answer:** A

Options: A) Influenza, chickenpox, and measles; B) Multiple sclerosis, Parkinson's disease, and Alzheimer's disease; C) Arthritis, osteoporosis, and fibromyalgia; D) Hypertension, heart disease, and diabetes; E) Chronic obstructive pulmonary disease (COPD), cystic fibrosis, and pleural effusion

Note: Correct option: A - Influenza, chickenpox, and measles

4. **Answer:** A

Options: A) the rate of mutations; B) the chance for variation in zygotes; C) the age of an organism; D) the potential for genetic disorders; E) the sequence of DNA bases

Note: Correct option: A - the rate of mutations

5. **Answer:** A

Options: A) The parathyroid glands produce insulin to regulate blood sugar levels; B) The parathyroid glands regulate blood pressure; C) The parathyroid glands control breathing rate; D) The parathyroid glands secrete enzymes that break down proteins; E) The parathyroid glands control the balance of electrolytes in the body

Note: Correct option: A - The parathyroid glands produce insulin to regulate blood sugar levels

True / False

6. **Answer:** True

Options: A) True; B) False

Note: This testing method may provide a useful strategy for conducting HIV surveillance in possible co-infected TB patients at peripheral centres. Since there is no in-

vestment on infrastructure, it may be possible for paramedical health professionals to carry out the test, particularly in areas with low HIV endemicity.

7. **Answer:** True

Options: A) True; B) False

Note: Patients who received a metal-backed Onlay tibial component obtained better postoperative mechanical alignment compared to those who received all-polyethylene Inlay prostheses. The thicker overall construct of Onlay prostheses appears to be an important determinant of postoperative alignment. Considering their higher survivorship rates and improved postoperative mechanical alignment, Onlay prostheses should be the first option when performing medial UKR.

8. **Answer:** True

Options: A) True; B) False

Note: The findings showed that acupuncture of voice-related acupoints could bring about improvement in vocal function and healing of vocal fold lesions.

9. **Answer:** True

Options: A) True; B) False

Note: Our results indicate that elevated cTnI levels are associated with higher risk for in-hospital mortality and complicated clinical course. Troponin I may play an important role for the risk assessment of patients with PE. The idea that an elevation in cTnI levels is a valuable parameter for the risk stratification of patients with PE needs to be examined in larger prospective studies.

10. **Answer:** True

Options: A) True; B) False

Note: These preliminary data support our hypothesis that recipient inflammation may affect RBC alloimmunization in humans; however, a more detailed understanding of the pathophysiologic association between inflammation and alloimmunization is required before definitive conclusions can be reached.

Long Form Response

11. **Answer:** Multiple alleles are three or more genes that control multiple traits. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** All individuals are identical genetically, and conjugation could not occur.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key